Practice Final Exam - Rod-Mass System: Answers and Derivations

This document contains all the derivations, calculations, and design values required for the practice final exam.

Part 1: Equations of Motion

1.1 Kinetic Energy (K)

The mass is a point mass at distance ℓ. Velocity $v = \ell\dot{\theta}$.

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$$

1.2 Potential Energy (P)

Includes gravity and the nonlinear spring. - $P_{gravity} = mgh = mg\ell\sin\theta$ (assuming $h = 0$ at horizontal) - $P_{spring} = \frac{1}{2}k_1\theta^2 + \frac{1}{4}k_2\theta^4$

$$P = mg\ell\sin\theta + \frac{1}{2}k_1\theta^2 + \frac{1}{4}k_2\theta^4$$

1.3 Lagrangian ($L = K - P$)

$$L = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell\sin\theta - \frac{1}{2}k_1\theta^2 - \frac{1}{4}k_2\theta^4$$

1.4 Generalized Forces (Q)

Includes input torque τ and viscous friction $-b\dot{\theta}$.

$$Q = \tau - b\dot{\theta}$$

1.5 Euler-Lagrange Equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}}\right) - \frac{\partial L}{\partial\theta} = Q$$

- $\frac{\partial L}{\partial\dot{\theta}} = m\ell^2\dot{\theta} \implies \frac{d}{dt}(\cdot) = m\ell^2\ddot{\theta}$ - $\frac{\partial L}{\partial\theta} = -mg\ell\cos\theta - k_1\theta - k_2\theta^3$

$$m\ell^2\ddot{\theta} + mg\ell\cos\theta + k_1\theta + k_2\theta^3 = \tau - b\dot{\theta}$$

$$\ddot{\theta} = \frac{1}{m\ell^2}(\tau - b\dot{\theta} - mg\ell\cos\theta - k_1\theta - k_2\theta^3)$$

Part 2: Design Models

2.1 Equilibrium Torque (τ_e)

At rest ($\dot{\theta} = 0, \ddot{\theta} = 0$) at angle θ_e:

$$\tau_e = mg\ell \cos\theta_e + k_1\theta_e + k_2\theta_e^3$$

For $\theta_e = 0$:

$$\tau_e = mg\ell = (0.1)(9.8)(0.25) = 0.245 \text{ N-m}$$

2.3 Linearization

Linearizing around $\theta_e = 0, \tau_e = mg\ell$:

$$\Delta\ddot{\theta} + \frac{b}{m\ell^2}\Delta\dot{\theta} + \frac{k_1}{m\ell^2}\Delta\theta = \frac{1}{m\ell^2}\Delta\tau$$

2.4 Transfer Function

$$P(s) = \frac{\Theta(s)}{T(s)} = \frac{1/m\ell^2}{s^2 + \frac{b}{m\ell^2}s + \frac{k_1}{m\ell^2}}$$

With nominal parameters ($m\ell^2 = 0.00625$):

$$P(s) = \frac{160}{s^2 + 16s + 3.2}$$

2.5 State Space Model

$$A = \begin{bmatrix} 0 & 1 \\ -3.2 & -16 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 160 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Part 3: PID Control

3.3 Proportional Gain (k_p) for Saturation

If $\tau_{max} = 3.0$ and we saturate at $\theta_e = 20°$ (0.349 rad):

$$\Delta\tau = k_p\Delta\theta \implies 3.0 - 0.245 = k_p(0.349) \implies k_p \approx 7.9$$

3.4 PD Design for Poles

Desired: $\Delta_{cl}(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$. Actual: $s^2 + (\frac{b+kd}{m\ell^2})s + (\frac{k_1+kp}{m\ell^2})$. For $\zeta = 0.9$ and ω_n chosen for performance: - $\omega_n = \sqrt{(k_1 + k_p)/m\ell^2}$ - $k_d = 2\zeta\omega_n m\ell^2 - b$

Part 4: State Space Control

4.1 SSI Gains

Gains $K = [k_\theta, k_{\dot{\theta}}]$ and k_i placed such that poles are at e.g., $[-5, -5.1, -10]$. Typical values (from `ssi_controller.py`): - $K = [0.039, 0.015]$ (Example, depends on pole locations) - $k_i = 0.0625$

4.3 Observer Gains (L)

Poles placed 5x faster than controller poles.

$$L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$$

4.4 Disturbance Observer (L_2)

Poles for state and disturbance $[x, d]$ placed at e.g., $[-25, -26, -27]$. Provides rejection of torque disturbances like the 0.5 N-m offset in Part 4.5.

4.7 LQR Optimization

$Q = \mathrm{diag}([10, 1, 100])$ (Penalize position, velocity, and integral error). $R = [0.1]$ (Penalize torque). Optimal K, k_i derived from Riccati equation.